

Similarity Functions

Neo4j Graph Data Science Library

Definitions

The Neo4j GDS library provides a set of measures that can be used to calculate similarity between two arrays p_s, p_t of numbers.

The similarity functions can be classified into two groups.

- The first is **categorical measures** which treat the arrays as sets and calculate similarity based on the intersection between the two sets.
- The second is **numerical measures** which compute similarity based on how close the numbers at each position are to each other.

Similarity functions

Six built-in Cypher functions to compute similarity between two arrays of numbers:

Similarity Function name	Formula	Type	Value range
<code>gds.similarity.jaccard</code>	$J(p_s, p_t) = \frac{ p_s \cap p_t }{ p_s \cup p_t }$	Categorical	[0, 1]
<code>gds.similarity.overlap</code>	$O(p_s, p_t) = \frac{ p_s \cap p_t }{\min(p_s , p_t)}$	Categorical	[0, 1]
<code>gds.similarity.cosine</code>	$\cosine(p_s, p_t) = \frac{\sum_i p_s(i) \cdot p_t(i)}{\sqrt{\sum_i p_s(i)^2} \cdot \sqrt{\sum_i p_t(i)^2}}$	Numerical	[-1, 1]
<code>gds.similarity.pearson</code>	$pearson(p_s, p_t) = \frac{\sum_i (p_s(i) - \bar{p}_s) \cdot (p_t(i) - \bar{p}_t)}{\sqrt{\sum_i (p_s(i) - \bar{p}_s)^2} \cdot \sqrt{\sum_i (p_t(i) - \bar{p}_t)^2}}$	Numerical	[-1, 1]
<code>gds.similarity.euclideanDistance</code>	$ED(p_s, p_t) = \sqrt{\sum_i (p_s(i) - p_t(i))^2}$	Numerical	[0, ∞)
<code>gds.similarity.euclidean</code>	$euclidean(p_s, p_t) = \frac{1}{1 + \sqrt{\sum_i (p_s(i) - p_t(i))^2}}$	Numerical	(0, 1]

Jaccard Similarity

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

```
RETURN gds.similarity.jaccard(  
    [1.0, 5.0, 3.0, 6.7],  
    [5.0, 2.5, 3.1, 9.0]  
) AS jaccardSimilarity
```

Result: **0.1429**

Overlap Coefficient

$$O(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)}$$

```
RETURN gds.similarity.overlap(  
    [1.0, 5.0, 3.0, 6.7],  
    [5.0, 2.5, 3.1, 9.0]  
) AS overlapSimilarity
```

Result: 0.25

Cosine Similarity

$$\cos(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

```
RETURN gds.similarity.cosine(  
    [1.0, 5.0, 3.0, 6.7],  
    [5.0, 2.5, 3.1, 9.0]  
) AS cosineSimilarity
```

Result: **0.8828**

Range: [-1, 1] — direction of vectors, not magnitude

Pearson Correlation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

```
RETURN gds.similarity.pearson(  
    [1.0, 5.0, 3.0, 6.7],  
    [5.0, 2.5, 3.1, 9.0]  
) AS pearsonSimilarity
```

Result: **0.4683**

Measures linear correlation between two vectors.

Euclidean Distance

$$d = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$$

```
RETURN gds.similarity.euclideanDistance(  
  [1.0, 5.0, 3.0, 6.7],  
  [5.0, 2.5, 3.1, 9.0]  
) AS euclideanDistance
```

Result: **5.2488**

Range: $[0, \infty)$ — lower means more similar

Euclidean Similarity

$$score = \frac{1}{1 + d}$$

```
RETURN gds.similarity.euclidean(  
    [1.0, 5.0, 3.0, 6.7],  
    [5.0, 2.5, 3.1, 9.0]  
) AS euclideanSimilarity
```

Result: **0.1600**

Converts Euclidean distance to a similarity score in (0, 1].

Null Handling (1/2)

Categorical (Jaccard, Overlap):

- `null` values are **excluded** from the set

```
// [1.0, 3.0] vs [1.0, 2.0, 3.0]  
RETURN gds.similarity.jaccard(  
  [1.0, null, 3.0],  
  [1.0, 2.0, 3.0]  
) AS jaccardSimilarity -- 0.6667
```

Null Handling (2/2)

Numerical (Cosine, Pearson, Euclidean):

- `null` values are replaced with `0.0`

```
// treated as [1.0, 0.0, 3.0] vs [1.0, 2.0, 3.0]
RETURN gds.similarity.cosine(
  [1.0, null, 3.0],
  [1.0, 2.0, 3.0]
) AS cosineSimilarity    -- 0.8452
```

Function Summary

Function	Type	Range	Use Case
jaccard	Categorical	[0, 1]	Set overlap (binary features)
overlap	Categorical	[0, 1]	Asymmetric set similarity
cosine	Numerical	[-1, 1]	Vector embeddings, text
pearson	Numerical	[-1, 1]	Linear correlation
euclideanDistance	Numerical	[0, ∞)	Raw geometric distance
euclidean	Numerical	(0, 1]	Normalized distance score

Key Takeaways

- Lightweight **Cypher functions** — no graph projection needed
- Two categories: **categorical** (set-based) and **numerical** (position-based)
- Input: two arrays of equal length
- `null` handling differs by category
- Used as building blocks in Node Similarity, KNN, and ML pipelines

References

- [Similarity Functions Documentation](#)

